FUNGSI TURUNAN VEKTOR NABLA OPERATOR, GRADIEN DAN ROTASI (Curl) DALAM OPERASI VEKTOR

FUNGSI VEKTOR

Jika sembarang nilai skalar t dikaitkan dengan suatu vektor **A**, maka **A** bisa dinyatakan sebagai fungsi vektor dari t atau $\mathbf{A}(t)$, yaitu suatu vektor yang komponen-komponennya merupakan fungsi dari nilai skalar t.

Dalam R^2, fungsi vektor $\mathbf{A}(t)$ biasa ditulis dengan,

$$\boxed{\mathbf{A}(t) = A_1(t)\mathbf{i} + A_2(t)\mathbf{j}}$$

dalam R^3, fungsi vektor $\mathbf{A}(t)$ ditulis dengan,

$$\boxed{\mathbf{A}(t) = A_1(t)\mathbf{i} + A_2(t)\mathbf{j} + A_3(t)\mathbf{k}}$$

FUNGSI VEKTOR

Konsep fungsi vektor ini bisa diperluas, jika sembarang titik (x,y,z) di R^3 dikaitkan dengan suatu vektor **A**, maka **A** bisa dinyatakan dalam bentuk fungsi vektor sebagai berikut:

$$\mathbf{A}(x, y, z) = A_1(x, y, z)\mathbf{i} + A_2(x, y, z)\mathbf{j} + A_3(x, y, z)\mathbf{k}$$

$$\mathbf{F} = xyz^2\mathbf{i} + yz^2\mathbf{j} + 2xy^2\mathbf{k},$$

Definisi Turunan Vektor

$\mathbf{A}(t)$ adalah sebuah fungsi vektor yang bergantung pada sebuah variabel t, didefinisikan turunan dari$\mathbf{A}(t)$ sebagai berikut:

$$\frac{d\mathbf{A}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{A}(t + \Delta t) - \mathbf{A}(t)}{\Delta t}$$

Jika fungsi vektor $\mathbf{A}(t) = A_1(t)\mathbf{i} + A_2(t)\mathbf{j} + A_3(t)\mathbf{k}$ dengan fungsi skalar-fungsi skalar $A_1(t)$, $A_2(t)$, dan $A_3(t)$ dapat diferensialkan terhadap variabel t, maka $\mathbf{A}(t)$ mempunyai turunan variabel terhadap t yang dirumuskan sebagai berikut:

$$\frac{d\mathbf{A}}{dt} = \frac{dA_1}{dt}\mathbf{i} + \frac{dA_2}{dt}\mathbf{j} + \frac{dA_3}{dt}\mathbf{k}$$

Sifat Turunan Biasa Fungsi Vektor

Jika **A**, **B**, dan **C** adalah fungsi-fungsi vektor dari sebuah skalar t yang diferensiabel dan ϕ sebuah fungsi skalar dari t yang diferensiabel, maka

i. $\frac{d}{dt}(\mathbf{A}+\mathbf{B}) = \frac{d\mathbf{A}}{dt} + \frac{d\mathbf{B}}{dt}$

ii. $\frac{d}{dt}(\mathbf{A}.\mathbf{B}) = \mathbf{A}.\frac{d\mathbf{B}}{dt} + \frac{d\mathbf{A}}{dt}.\mathbf{B}$

iii. $\frac{d}{dt}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\frac{d\mathbf{B}}{dt} + \frac{d\mathbf{A}}{dt}\times\mathbf{B}$

iv. $\frac{d}{dt}(\phi\mathbf{A}) = \phi\frac{d\mathbf{A}}{dt} + \frac{d\phi}{dt}\mathbf{A}$

v. $\frac{d}{dt}(\mathbf{A}.\mathbf{B}\times\mathbf{C}) = \mathbf{A}.\mathbf{B}\times\frac{d\mathbf{C}}{dt} + \mathbf{A}.\frac{d\mathbf{B}}{dt}\times\mathbf{C} + \frac{d\mathbf{A}}{dt}.\mathbf{B}\times\mathbf{C}$

vi. $\frac{d}{dt}\{\mathbf{A}\times(\mathbf{B}\times\mathbf{C})\} = \mathbf{A}\times\left(\mathbf{B}\times\frac{d\mathbf{C}}{dt}\right) + \mathbf{A}\times\left(\frac{d\mathbf{B}}{dt}\times\mathbf{C}\right) + \frac{d\mathbf{A}}{dt}\times(\mathbf{B}\times\mathbf{C})$

Contoh 1

Jika $\mathbf{f}(t) = e^{\sin(t^2+2t)}\mathbf{i} + \ln(t^2+2t)\mathbf{j} + 4t^3\mathbf{k}$, tentukan $\frac{d\mathbf{f}}{dt}$

Penyelesaian

$$\frac{d\mathbf{f}}{dt} = \frac{df_1}{dt}\mathbf{i} + \frac{df_2}{dt}\mathbf{j} + \frac{df_3}{dt}\mathbf{k}$$

$$= \frac{d\left(e^{\sin(t^2+2t)}\right)}{dt}\mathbf{i} + \frac{d(\ln(t^2+2t))}{dt}\mathbf{j} + \frac{d(4t^3)}{dt}\mathbf{k}$$

$$= (2t+2)\cos(t^2+2t)e^{\sin(t^2+2t)}\mathbf{i} + \frac{2t+2}{t^2+2t}\mathbf{j} + 12t^2\mathbf{k}$$

Jika $\mathbf{A} = (t^2 + 2t)\mathbf{i} + 2t\mathbf{j} + t^3\mathbf{k}$ dan $\mathbf{B} = 2t\mathbf{i} + \sin t^2\,\mathbf{j} + 4t\mathbf{k}$. Tentukan $\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B})$ di $t = 0$

$$\begin{aligned}\mathbf{A}\cdot\mathbf{B} &= A_1B_1 + A_2B_2 + A_3B_3\\ &= (t^2 + 2t)2t + 2t\sin t^2 + 4t^4\\ &= 2t^3 + 4t^2 + 2t\sin t^2 + 4t^4\end{aligned}$$

$$\begin{aligned}\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) &= \frac{d}{dt}[2t^3 + 4t^2 + 2t\sin t^2 + 4t^4]\\ &= 6t^2 + 8t + 4t^2\cos t^2 + 2\sin t^2 + 16t^3\end{aligned}$$

pada saat $t = 0$, maka

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = 0$$

Cara 2 (menggunakan sifat turunan)

$$\begin{aligned}
\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) &= \mathbf{A}\cdot\frac{d\mathbf{B}}{dt}+\frac{d\mathbf{A}}{dt}\cdot\mathbf{B} \\
&= [(t^2+2t)\mathbf{i}+2t\mathbf{j}+t^3\mathbf{k}]\cdot(2\mathbf{i}+2t\cos t^2\,\mathbf{j}+4\mathbf{k}) \\
&\quad + [(2t+2)\mathbf{i}+2\mathbf{j}+3t^2\mathbf{k}]\cdot(2t\mathbf{i}+\sin t^2\,\mathbf{j}+4t\mathbf{k}) \\
&= (t^2+2t)2+4t^2\cos t^2+4t^3+(2t+2)2t+2\sin t^2+12t^3 \\
&= 16t^3+6t^2+8t+4t^2 \quad \cos t^2+2\sin t^2
\end{aligned}$$

pada saat $t=0$, maka

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = 0$$

Jika $\mathbf{r}(t) = (3t^2 - 1)\mathbf{i} + (t^2 + 1)\mathbf{j} + t^2\mathbf{k}$, tentukan vektor singgung satuan pada titik $t = 1$.

$$\frac{d\mathbf{r}}{dt} = \frac{d}{dt}[(3t^2 - 1)\mathbf{i} + (t^2 + 1)\mathbf{j} + t^2\mathbf{k}]$$

$$= 6t\mathbf{i} + 2t\mathbf{j} + 2t\mathbf{k}$$

$$\left|\frac{d\mathbf{r}}{dt}\right| = \sqrt{(6t)^2 + (2t)^2 + (2t)^2} = \sqrt{44t^2} = t\sqrt{44}$$

$$\mathbf{T} = \frac{6t\mathbf{i} + 2t\mathbf{j} + 2t\mathbf{k}}{t\sqrt{44}}$$

Saat $t = 1$, maka $\mathbf{T} = \frac{6\mathbf{i}+2\mathbf{j}+2\mathbf{k}}{\sqrt{44}}$

TURUNAN VEKTOR PARSIAL

Turunan parsial untuk fungsi vektor dua variabel atau lebih, prinsipnya sama dengan definisi turunan fungsi vektor satu variabel, dimana semua variabel dianggap konstan, kecuali satu yaitu variabel terhadap apa fungsi vektor itu diturunkan.

Misalkan A adalah sebuah fungsi vektor yang tergantung kepada variabel skalar x, y, dan z maka kita tuliskan A = A (x, y, z). Jika limitnya ada, ketiga turunan parsialnya didefinisikan masing-masing sebagai berikut:

$$\frac{\partial \mathbf{A}}{\partial x} = \lim_{\Delta x \to 0} \frac{\mathbf{A}(x + \Delta x, y, z) - \mathbf{A}(x, y, z)}{\Delta x}$$

$$\frac{\partial \mathbf{A}}{\partial y} = \lim_{\Delta y \to 0} \frac{\mathbf{A}(x, y + \Delta y, z) - \mathbf{A}(x, y, z)}{\Delta y}$$

$$\frac{\partial \mathbf{A}}{\partial z} = \lim_{\Delta z \to 0} \frac{\mathbf{A}(x, y, z + \Delta z) - \mathbf{A}(x, y, z)}{\Delta z}$$

Jika fungsi vektor $A(x,y,z) = A_1(x,y,z)i + A_2(x,y,z)j + A_3(x,y,z)k$ dengan fungsi skalarnya $A_1(x,y,z), A_2(x,y,z)$, dan $A_3(x,y,z)$ mempunyai turunan parsial terhadap variabel x, y, dan z maka A (x, y, z) juga mempunyai turunan variabel terhadap x, y, dan z yang dirumuskan sebagai berikut:

$$\frac{\partial \mathbf{A}}{\partial x} = \frac{\partial A_1}{\partial x}\mathbf{i} + \frac{\partial A_2}{\partial x}\mathbf{j} + \frac{\partial A_3}{\partial x}\mathbf{k}$$

$$\frac{\partial \mathbf{A}}{\partial y} = \frac{\partial A_1}{\partial y}\mathbf{i} + \frac{\partial A_2}{\partial y}\mathbf{j} + \frac{\partial A_3}{\partial y}\mathbf{k}$$

$$\frac{\partial \mathbf{A}}{\partial z} = \frac{\partial A_1}{\partial z}\mathbf{i} + \frac{\partial A_2}{\partial z}\mathbf{j} + \frac{\partial A_3}{\partial z}\mathbf{k}$$

SIFAT-SIFAT TURUNAN PARSIAL

Berikut sifat-sifat turunan parsial:

Misalkan **A** dan **B** adalah fungsi-fungsi vektor dan ϕ adalah fungsi skalar x, y, dan z dan dapat dideferensialkan terhadap ketiga variabel tersebut, maka berlaku

i. $\frac{\partial}{\partial x}(\mathbf{A}+\mathbf{B}) = \frac{\partial \mathbf{A}}{\partial x} + \frac{\partial \mathbf{B}}{\partial x}$

ii. $\frac{\partial}{\partial x}(\phi\mathbf{A}) = \phi\frac{\partial \mathbf{A}}{\partial x} + \frac{\partial \phi}{\partial x}\mathbf{A}$

iii. $\frac{\partial}{\partial x}(\mathbf{A}.\mathbf{B}) = \mathbf{A}.\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}.\mathbf{B}$

iv. $\frac{\partial}{\partial x}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}\times\mathbf{B}$

v. $\frac{\partial^2}{\partial y\partial x}(\mathbf{A}.\mathbf{B}) = \frac{\partial}{\partial y}\left\{\frac{\partial}{\partial x}(\mathbf{A}.\mathbf{B})\right\} = \frac{\partial}{\partial y}\left\{\mathbf{A}.\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}.\mathbf{B}\right\}$

$$= \mathbf{A}.\frac{\partial^2 \mathbf{B}}{\partial y\partial x} + \frac{\partial \mathbf{A}}{\partial y}.\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}.\frac{\partial \mathbf{B}}{\partial y} + \frac{\partial^2 \mathbf{A}}{\partial y\partial x}.\mathbf{B}$$

ATURAN RANTAI UNTUK MENENTUKAN TURUNAN VEKTOR

Misalkan $F = F(x, y, z)$ adalah fungsi vektor yang dapat dideferensialkan terhadap variabel x, y, dan z, dimana $x = x(s, t, u)$, $y = y(s, t, u)$, dan $z = z(s, t, u)$ adalah fungsi-fungsi skalar yang dapat dideferensialkan terhadap variabel s, t, dan u, maka bentuk fungsi tersusun F dapat dituliskan dengan

$$\boxed{F = F[x(s, t, u), y(s, t, u), z(s, t, u)]}$$

Turunan parsial F terhadap variabel s,t , dan u dapat diberikan sebagai berikut:

$$\boxed{\begin{aligned}
\frac{\partial \mathbf{F}}{\partial s} &= \frac{\partial \mathbf{F}}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial \mathbf{F}}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial \mathbf{F}}{\partial z}\frac{\partial z}{\partial s} \\
\frac{\partial \mathbf{F}}{\partial t} &= \frac{\partial \mathbf{F}}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial \mathbf{F}}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial \mathbf{F}}{\partial z}\frac{\partial z}{\partial t} \\
\frac{\partial \mathbf{F}}{\partial u} &= \frac{\partial \mathbf{F}}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial \mathbf{F}}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial \mathbf{F}}{\partial z}\frac{\partial z}{\partial u}
\end{aligned}}$$

CONTOH SOAL

Jika $\mathbf{F} = xyz^2\mathbf{i} + yz^2\mathbf{j} + 2xy^2\mathbf{k}$, tentukanlah (a)$\frac{\partial \mathbf{F}}{\partial x}$, (b)$\frac{\partial \mathbf{F}}{\partial y}$, (c)$\frac{\partial \mathbf{F}}{\partial z}$

Penyelesaian

$$
\begin{aligned}
\text{(a)}\ \frac{\partial \mathbf{F}}{\partial x} &= \frac{\partial}{\partial x}(xyz^2\mathbf{i} + yz^2\mathbf{j} + 2xy^2\mathbf{k}) \\
&= yz^2\mathbf{i} + 0 + 2y^2\mathbf{k} \\
&= yz^2\mathbf{i} + 2y^2\mathbf{k} \\
\text{(b)}\ \frac{\partial \mathbf{F}}{\partial y} &= \frac{\partial}{\partial y}(xyz^2\mathbf{i} + yz^2\mathbf{j} + 2xy^2\mathbf{k}) \\
&= xz^2\mathbf{i} + z^2\mathbf{j} + 4xy\mathbf{k} \\
\text{(c)}\ \frac{\partial \mathbf{F}}{\partial z} &= \frac{\partial}{\partial z}(xyz^2\mathbf{i} + yz^2\mathbf{j} + 2xy^2\mathbf{k}) \\
&= 2xyz\mathbf{i} + 2yz\mathbf{j} + 0 \\
&= 2xyz\mathbf{i} + 2yz\mathbf{j}
\end{aligned}
$$

LATIHAN SOAL 2. TURUNAN PARSIAL

1. Jika $F = \sin xy^2 z\boldsymbol{i} + 2yz\boldsymbol{j} + z^2\boldsymbol{k}$. Tentukan $\frac{\partial^2 F}{\partial x^2}$
2. Jika $W = (x^2 + y^2)\boldsymbol{i} + z^2\boldsymbol{j} + xy\boldsymbol{k}, x = st, y = s - t, z = s + 2t$. Tentukan $\frac{\partial W}{\partial t}$
3. Jika $A = \cos xy\boldsymbol{i} + (3xy - 2x^2)\boldsymbol{j} - (3x + 2y)\boldsymbol{k}$, carilah $\frac{\partial A}{\partial x}, \frac{\partial A}{\partial y}, \frac{\partial^2 A}{\partial x^2}, \frac{\partial^2 A}{\partial y^2}, \frac{\partial^2 A}{\partial x \partial y}, \frac{\partial^2 A}{\partial y \partial x}$
4. Jika $A = x^2 yz\boldsymbol{i} - 2xz^3\boldsymbol{j} + xz^2\boldsymbol{k}$ dan $B = 2z\boldsymbol{i} + y\boldsymbol{j} + x^2\boldsymbol{k}$, carilah $\frac{\partial^2}{\partial x \partial y}(A \times B)$ di titik (1, 0, -2)
5. Misalkan $F = e^{xyz}\boldsymbol{i}$ dimana $x = stu^2, y = st^2u + us$, dan $z = st + tu$. Tentukan (a) $\frac{\partial F}{\partial s}$, (b) $\frac{\partial F}{\partial t}$, (c) $\frac{\partial F}{\partial u}$

Gradien, Divergensi & Curl

Operator Del atau Nabla

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$

Divergensi

$$\nabla \bullet A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) \bullet \left(A_1 i + A_2 j + A_3 k \right)$$

$$= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z}$$

Curl

$$\nabla \times A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) \times \left(A_1 i + A_2 j + A_3 k \right)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) i + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) j + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) k$$

Gadien, Divergensi, dan Curl

OPERATOR DEL

Operator del merupakan operator pada diferensial vektor yang disimbolkan dengan (nabla), yang didefinisikan dalam bentuk turunan parsial, yaitu:

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$

Operator del ini bermanfaat untuk mencari gradien, divergensi, dan curl.

Gradien

- Misalkan terdefinisi dan diferensiabel pada setiap titik (*x,y,z*) dalam ruang , maka gradien atau grad atau didefinisikan oleh :

$$\nabla\phi = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right)\phi$$

$$= \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

Contoh

Jika $\phi = 2xz^4 - x^2y$, carilah $\nabla\phi$ dan $|\nabla\phi|$ pada titik (2, -2, 1)

Penyelesaian

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

$$= \frac{\partial(2xz^4 - x^2y)}{\partial x}\mathbf{i} + \frac{\partial(2xz^4 - x^2y)}{\partial y}\mathbf{j} + \frac{\partial(2xz^4 - x^2y)}{\partial z}\mathbf{k}$$

$$= (2z^4 - 2xy)\boldsymbol{i} - x^2\mathbf{j} + 8xz^3\mathbf{k}$$

$$\nabla\phi(2,-2,1) = 10\mathbf{i} - \mathbf{4j} + 16\mathbf{k}$$

$$|\nabla\phi| = \sqrt{10^2 + (-4)^2 + 16^2} = 2\sqrt{93}$$

Sifat-Sifat Gradien

Sifat-sifat gradien

Misalkan $\phi(x, y, z)$ dan $\psi(x, y, z)$ adalah fungsi-fungsi skalar yang diferensiabel pada setiap titik (x, y, z) dan c adalah bilangan real, maka berlaku:

i. $\nabla(\phi + \psi) = \nabla\phi + \nabla\psi$

ii. $\nabla(c\phi) = c(\nabla\phi)$

iii. $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$

Contoh

Jika $\phi = 2xz^4 - x^2y$, carilah $\nabla\phi$ dan $|\nabla\phi|$ pada titik (2, -2, 1)

Penyelesaian

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

$$= \frac{\partial(2xz^4 - x^2y)}{\partial x}\mathbf{i} + \frac{\partial(2xz^4 - x^2y)}{\partial y}\mathbf{j} + \frac{\partial(2xz^4 - x^2y)}{\partial z}\mathbf{k}$$

$$= (2z^4 - 2xy)\boldsymbol{i} - x^2\mathbf{j} + 8xz^3\mathbf{k}$$

$$\nabla\phi(2,-2,1) = 10\mathbf{i} - \mathbf{4j} + 16\mathbf{k}$$

$$|\nabla\phi| = \sqrt{10^2 + (-4)^2 + 16^2} = 2\sqrt{93}$$

Latihan Soal:

Latihan 1

Misalkan $\phi = (x, y, z) = e^{xyz}$, tentukanlah:

a. $\nabla\phi$ pada titik (1,2,1)

b. $|\nabla\phi|$ pada titik (1,2,1)

c. $\mathbf{n}$, jika $\mathbf{n}$ vektor satuan dari $\nabla\phi$ pada titik (1,2,1)

Divergensi

- Misalkan vektor V(x, y, *z)* = terdefinisikan dan diferensiabel pada setiap titik (*x,y,z*). Divergensi dari V atau div V , didefinisikan oleh:

$$\nabla \cdot \mathbf{V} = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right) \cdot \left(V_1\mathbf{i} + V_2\mathbf{j} + V_3\mathbf{k}\right)$$

$$= \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

Sifat-Sifat Divergensi

Sifat-sifat divergensi:

Misalkan $\mathbf{F}(x, y, z)$ dan $\mathbf{G}(x, y, z)$ adalah vektor-vektor yang kontinu dan diferensiabel terhadap x, y, dan z, $\phi(x, y, z)$ adalah fungsi skalar yang kontinu dan diferensiabel terhadap x, y, dan z, serta a dan b adalah bilangan real, maka berlaku

i. $\nabla \cdot (a\mathbf{F} + b\mathbf{G}) = a\nabla \cdot \mathbf{F} + b\nabla \cdot \mathbf{G}$

ii. $\nabla \cdot (\phi\mathbf{F}) = \phi(\nabla \cdot \mathbf{F}) + (\nabla\phi) \cdot \mathbf{F}$

iii. $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G})$

iv. $\nabla \cdot (\nabla \times \mathbf{F}) = 0$

Contoh

Jika $\mathbf{A} = 3xyz^2\mathbf{i} + 2xy^3\mathbf{j} - x^2yz\mathbf{k}$ dan $\phi = 3x^2 - yz$.

Carilah (a)$\nabla \cdot \mathbf{A}$, (b)$\mathbf{A} \cdot \nabla\phi$ di titik (1, -1, 1)

Penyelesaian

(a) $\nabla \cdot \mathbf{A} = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right).(3xyz^2\mathbf{i} + 2xy^3\mathbf{j} - x^2yz\mathbf{k})$

$$= 3yz^2 + 6xy^2 - x^2y$$

$$\nabla \cdot \mathbf{A}(1,-1,1) = 3.(-1).1^2 + 6.1.(-1)^2 - 1^2.(-1) = 4$$

(b) $\mathbf{A} \cdot \nabla\phi = (3xyz^2\mathbf{i} + 2xy^3\mathbf{j} - x^2yz\mathbf{k}) \cdot \left(\frac{\partial(3x^2-yz)}{\partial x}\mathbf{i} + \frac{\partial(3x^2-yz)}{\partial y}\mathbf{j} + \frac{\partial(3x^2-yz)}{\partial z}\mathbf{k}\right)$

$$= (3xyz^2\mathbf{i} + 2xy^3\mathbf{j} - x^2yz\mathbf{k}) \cdot (6x\mathbf{i} - z\mathbf{j} - y\mathbf{k})$$

$$= 18x^2yz^2 - 2xy^3z + x^2y^2z$$

$$\mathbf{A} \cdot \nabla\phi(1,-1,1) = 18 \cdot 1^2 \cdot (-1) \cdot 1^2 - 2 \cdot 1 \cdot (-1)^3 \cdot 1 + 1^2 \cdot (-1)^2 \cdot 1$$

$$= -15$$

Curl

Jika vektor terdefinisi dan diferensiabel pada setiap titik (*x,y,z*), maka curl atau rotasi dari V,dituliskan curl V atau rot V , didefinisikan oleh:

$$\nabla \times \mathbf{V} = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right) \times (V_1\mathbf{i} + V_2\mathbf{j} + V_3\mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_2 & V_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ V_1 & V_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ V_1 & V_2 \end{vmatrix} \mathbf{k}$$

$$\nabla \times \mathbf{V} = \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z}\right)\mathbf{i} + \left(\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x}\right)\mathbf{j} + \left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y}\right)\mathbf{k}$$

Sifat-sifat curl:

Misalkan $\mathbf{F}(x, y, z)$ dan $\mathbf{G}(x, y, z)$ adalah fungsi vektor-vektor yang kontinu dan diferensiabel terhadap x, y, dan z, $\phi(x, y, z)$ adalah fungsi skalar yang kontinu dan diferensiabel terhadap x, y, dan z, dan a adalah bilangan real, maka berlaku:

i. $\nabla \times (\mathbf{F} + \mathbf{G}) = (\nabla \times \mathbf{F}) + (\nabla \times \mathbf{G})$

ii. $\nabla \times a\mathbf{F} = a(\nabla \times \mathbf{F})$

iii. $\nabla \times \phi\mathbf{F} = (\nabla\phi) \times \mathbf{F} + \phi(\nabla \times \mathbf{F})$

iv. $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2\mathbf{F}$

v. $\nabla \times (\nabla\phi) = 0$

vi. $\nabla \times (\mathbf{F} \times \mathbf{G}) = (\mathbf{G} \cdot \nabla)\mathbf{F} - \mathbf{G}(\nabla \cdot \mathbf{F}) - (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{F}(\nabla \cdot \mathbf{G})$

Contoh soal

If $\mathbf{A} = xz^3\,\mathbf{i} - 2x^2yz\,\mathbf{j} + 2yz^4\,\mathbf{k}$, find $\nabla \times \mathbf{A}$ (or curl $\mathbf{A}$) at the point $(1,-1,1)$.

$$\nabla \times \mathbf{A} = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right) \times (xz^3\,\mathbf{i} - 2x^2yz\,\mathbf{j} + 2yz^4\,\mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}(2yz^4) - \frac{\partial}{\partial z}(-2x^2yz)\right]\mathbf{i} + \left[\frac{\partial}{\partial z}(xz^3) - \frac{\partial}{\partial x}(2yz^4)\right]\mathbf{j} + \left[\frac{\partial}{\partial x}(-2x^2yz) - \frac{\partial}{\partial y}(xz^3)\right]\mathbf{k}$$

$$= (2z^4 + 2x^2y)\,\mathbf{i} + 3xz^2\,\mathbf{j} - 4xyz\,\mathbf{k} = 3\mathbf{j} + 4\mathbf{k} \quad \text{at } (1,-1,1).$$

If $\mathbf{A} = x^2y\,\mathbf{i} - 2xz\,\mathbf{j} + 2yz\,\mathbf{k}$, find curl curl $\mathbf{A}$.

$$\text{curl curl } \mathbf{A} = \nabla \times (\nabla \times \mathbf{A})$$

$$= \nabla \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -2xz & 2yz \end{vmatrix} = \nabla \times \left[(2x + 2z)\mathbf{i} - (x^2 + 2z)\mathbf{k}\right]$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x + 2z & 0 & -x^2 - 2z \end{vmatrix} = (2x + 2)\mathbf{j}$$

Jika $\mathbf{Z} = 3x^2\mathbf{i} - y^2\mathbf{j}$, dengan $x = 2s + 7t$ dan $y = 5st$, tentukan $\frac{\partial z}{\partial t}$ dan nyatakan dalaam bentuk s dan t.

Jika $\mathbf{F} = \sin xy^2z\,\mathbf{i} + 2yz\mathbf{j} + z^2\mathbf{k}$. Tentukan $\frac{\partial^2 \mathbf{F}}{\partial x^2}$

Jika $\mathbf{W} = (x^2 + y^2)\mathbf{i} + z^2\mathbf{j} + xy\mathbf{k}$, $x = st, y = s - t, z = s + 2t$, tentukan $\frac{\partial \mathbf{W}}{\partial t}$

Jika $\mathbf{A} = x^2yz\mathbf{i} - 2xz^3\mathbf{j} + xz^2\mathbf{k}$ dan $\mathbf{B} = 2z\mathbf{i} + y\mathbf{j} + x^2\mathbf{k}$, carilah $\frac{\partial^2}{\partial x \partial y}(\mathbf{A} \times \mathbf{B})$ di titik (1,0,-2)

Sekian